

reliance on these models should be limited.”¹⁰ No one listened to them at that time.

However, we shouldn’t blame only the tools and not the workers who use them. One can make a case that these financial crises were due in large part to institutions using derivative models without exercising proper judgment or fully understanding their true risks. Derivative buyers of convoluted mortgage obligations should have listened to Nobel laureate George Akerlof who said, “If you are in a market and someone is trying to sell you something which you don’t understand, you should think they are selling you a lemon.”

The BS model had one interesting feature compared with other financial models. When people said, “Hey, finance isn’t rocket science,” financial economists could now answer, “Oh yes, it is!” Robert Merton (a Nobel laureate with a background in engineering mathematics) added Ito’s lemma to the BS model. Rocket scientists use Ito calculus to track the trajectory of rockets by breaking up the notion of continuous time into infinitely small pieces until it becomes a continuum. Not only did BS make economics look more like a science, it made it look like rocket science. Noble laureate Paul Krugman’s statement about economists could easily apply to financial economists: “As I see it, the economics profession went astray because economists as a group mistook beauty, clad in impressive looking mathematics, for truth.” In the words of Robert Heilbroner, “Mathematics has given economics rigor, but alas, also mortis.”¹¹

The biggest problem with the BS model is its inaccuracy when dealing with extreme price movement. The lognormal distribution of prices that underlies the BS model can underestimate the probability of risky events by a factor of 10.

Practitioners long ago replaced the BS model with a more realistic binomial pricing model, yet academics still count the BS model as one of the great discoveries of modern finance. In the words of Nobel laureate Robert Shiller, “Economics is usually some kind of story that we tell that we think approximates reality, but we can get carried away with our stories.”